

JEE Mains 2019 Chapter wise Question Bank

Application of Derivatives – Questions

Q1

The maximum volume (in cu.m) of the right circular cone having slant height 3 m is:

- (1) 6π (2) $3\sqrt{3}\pi$
 (3) $\frac{4}{3}\pi$ (4) $2\sqrt{3}\pi$

9 Jan Morning

Q2

Let $d \in \mathbb{R}$, and

$$A = \begin{bmatrix} -2 & 4+d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2 \sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix},$$

$\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$ is 8, then a value of d is:

- (1) -5 (2) -7
 (3) $2(\sqrt{2} + 1)$ (4) $2(\sqrt{2} + 2)$

10 Jan Morning

Q3

Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is:

- (1) $(0, \sqrt{2})$ (2) $(-\sqrt{2}, 0)$
 (3) $(\sqrt{2}, -\sqrt{2})$ (4) $(-\sqrt{2}, \sqrt{2})$

10 Jan Morning

Q4

The tangent to the curve, $y = xe^{x^2}$ passing through the point $(1, e)$ also passes through the point:

- (1) $(2, 3e)$ (2) $\left(\frac{4}{3}, 2e\right)$
 (3) $\left(\frac{5}{3}, 2e\right)$ (4) $(3, 6e)$

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Q5

A helicopter is flying along the curve given by $y - x^{3/2} = 7$, ($x \geq 0$). A soldier positioned at the point

$\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is:

- (1) $\frac{\sqrt{5}}{6}$ (2) $\frac{1}{3}\sqrt{\frac{7}{3}}$
 (3) $\frac{1}{6}\sqrt{\frac{7}{3}}$ (4) $\frac{1}{2}$

10 Jan Evening

Q6

The maximum value of the function $f(x) = 3x^3 - 18x^2 + 27x - 40$ on the set $S = \{x \in \mathbb{R} : x^2 + 30 \leq 11x\}$ is:

- (1) -122 (2) -222
 (3) 122 (4) 222

11 Jan Morning

Q7

Application of Derivatives

Let x, y be positive real numbers and m, n positive integers.

The maximum value of the expression $\frac{x^m y^n}{(1+x^{2m})(1+y^{2n})}$

is :

- (1) 1 (2) $\frac{1}{2}$
(3) $\frac{1}{4}$ (4) $\frac{m+n}{6mn}$

11 Jan Evening

Q8

Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$, $x \in \mathbf{R}$ where a, b

and d are non-zero real constants. Then :

- (1) f is an increasing function of x
(2) f is a decreasing function of x
(3) f' is not a continuous function of x
(4) f is neither increasing nor decreasing function of x

11 Jan Evening

Q9

The maximum area (in sq. units) of a rectangle having its base on the x -axis and its other two vertices on the parabola, $y = 12 - x^2$ such that the rectangle lies inside the parabola, is:

- (1) 36 (2) $20\sqrt{2}$
(3) 32 (4) $18\sqrt{3}$

12 Jan Morning

Q10

Let $P(4, -4)$ and $Q(9, 6)$ be two points on the parabola, $y^2 = 4x$ and let this X be any point arc POQ of this parabola, where O is vertex of the parabola, such that the area of ΔPXQ is maximum. Then this minimum area (in sq. units) is:

- (1) $\frac{75}{2}$ (2) $\frac{125}{4}$
(3) $\frac{625}{4}$ (4) $\frac{125}{2}$

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Q11

The maximum value of $3\cos\theta + 5\sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is:

- (1) $\sqrt{19}$ (2) $\frac{\sqrt{79}}{2}$
(3) $\sqrt{34}$ (4) $\sqrt{31}$

12 Jan Morning

Q12

If a curve passes through the point $(1, -2)$ and has slope of the tangent at any point (x, y) on it as $\frac{x^2 - 2y}{x}$, then the curve also passes through the point :

- (1) $(3, 0)$ (2) $(\sqrt{3}, 0)$ (3) $(-1, 2)$ (4) $(-\sqrt{2}, 1)$

12 Jan Evening

Q13

The tangent to the curve $y = x^2 - 5x + 5$, parallel to the line $2y = 4x + 1$, also passes through the point :

- (1) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (2) $\left(\frac{1}{8}, -7\right)$
(3) $\left(-\frac{1}{8}, 7\right)$ (4) $\left(\frac{1}{4}, \frac{7}{2}\right)$

12 Jan Evening

Q14

If the function f given by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in \mathbf{R}$ is increasing in $(0, 1]$ and decreasing in $[1, 5)$, then a root of the equation,

$$\frac{f(x) - 14}{(x-1)^2} = 0 (x \neq 1) \text{ is}$$

- (1) -7 (2) 5
(3) 7 (4) 6

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Application of Derivatives

Q15

The shortest distance between the line $y = x$ and the curve $y^2 = x - 2$ is :

- (1) 2 (2) $\frac{7}{8}$ (3) $\frac{7}{4\sqrt{2}}$ (4) $\frac{11}{4\sqrt{2}}$

8 April Morning

Q16

If S_1 and S_2 are respectively the sets of local minimum and local maximum points of the function,

$f(x) = 9x^4 + 12x^3 - 36x^2 + 25$, $x \in \mathbb{R}$, then :

- (1) $S_1 = \{-2\}$; $S_2 = \{0, 1\}$
(2) $S_1 = \{-2, 0\}$; $S_2 = \{1\}$
(3) $S_1 = \{-2, 1\}$; $S_2 = \{0\}$
(4) $S_1 = \{-1\}$; $S_2 = \{0, 2\}$

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Q17

Let $f: [0 : 2] \rightarrow \mathbb{R}$ be a twice differentiable function such that $f''(x) > 0$, for all $x \in (0, 2)$. If $\phi(x) = f(x) + f(2 - x)$, then ϕ is :

- (1) increasing on $(0, 1)$ and decreasing on $(1, 2)$.
(2) decreasing on $(0, 2)$
(3) decreasing on $(0, 1)$ and increasing on $(1, 2)$.
(4) increasing on $(0, 2)$

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Q18

Given that the slope of the tangent to a curve $y = y(x)$ at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the centre of

the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is :

- (1) $x \log_e |y| = 2(x - 1)$
(2) $x \log_e |y| = -2(x - 1)$
(3) $x^2 \log_e |y| = -2(x - 1)$
(4) $x \log_e |y| = x - 1$

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Q19

The height of a right circular cylinder of maximum volume inscribed in a sphere of radius 3 is :

- (1) $\sqrt{6}$ (2) $\frac{2}{3}\sqrt{3}$ (3) $2\sqrt{3}$ (4) $\sqrt{3}$

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Q20

If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$; then the set

$S = \{x \in \mathbb{R} : f(x) = f(0)\}$

contains exactly:

- (1) four irrational numbers.
(2) four rational numbers.
(3) two irrational and two rational numbers.
(4) two irrational and one rational number.

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Q21

If the tangent to the curve, $y = x^3 + ax - b$ at the point $(1, -5)$ is perpendicular to the line, $-x + y + 4 = 0$, then which one of the following points lies on the curve?

- (1) $(-2, 1)$ (2) $(-2, 2)$
(3) $(2, -1)$ (4) $(2, -2)$

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Q22

Let S be the set of all values of x for which the tangent to the curve $y = f(x) = x^3 - x^2 - 2x$ at (x, y) is parallel to the line segment joining the points $(1, f(1))$ and $(-1, f(-1))$, then S is equal to:

- (1) $\left\{\frac{1}{3}, 1\right\}$ (2) $\left\{-\frac{1}{3}, -1\right\}$
(3) $\left\{\frac{1}{3}, -1\right\}$ (4) $\left\{-\frac{1}{3}, 1\right\}$

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Q23

A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is

poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10m; is:

- (1) $1/15 \pi$ (2) $1/10 \pi$ (3) $2/\pi$ (4) $1/5 \pi$

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Application of Derivatives

Q24

Let $f(x) = e^x - x$ and $g(x) = x^2 - x$, $\forall x \in \mathbf{R}$. Then the set of all $x \in \mathbf{R}$, where the function $h(x) = (f \circ g)(x)$ is increasing, is :

- (1) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (2) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
(3) $[0, \infty)$ (4) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$

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Q25

If the tangent to the curve $y = \frac{x}{x^2 - 3}$, $x \in \mathbf{R}$, ($x \neq \pm\sqrt{3}$), at

a point (α, β) is parallel to the line $2x + 6y - 11 = 0$, then :

- (1) $|6\alpha + 2\beta| = 19$ (2) $|6\alpha + 2\beta| = 9$
(3) $|2\alpha + 6\beta| = 19$ (4) $|2\alpha + 6\beta| = 11$

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Q26

A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is :

- (1) $\frac{1}{18\pi}$ (2) $\frac{1}{36\pi}$
(3) $\frac{5}{6\pi}$ (4) $\frac{1}{9\pi}$

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Q27

Let $a_1, a_2, a_3, \dots$ be an A. P. with $a_6 = 2$. Then the common difference of this A.P., which maximises the product $a_1 a_4 a_5$, is :

- (1) $\frac{3}{2}$ (2) $\frac{8}{5}$ (3) $\frac{6}{5}$ (4) $\frac{2}{3}$

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Q28

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If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval $[0, 3]$ and M is the maximum value of f in $[0, 3]$ when $k = m$, then the ordered pair (m, M) is equal to :

- (1) $(4, 3\sqrt{2})$ (2) $(4, 3\sqrt{3})$ (3) $(3, 3\sqrt{3})$ (4) $(5, 3\sqrt{6})$

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Q29

The equation of a common tangent to the curves, $y^2 = 16x$ and $xy = -4$, is :

- (1) $x - y + 4 = 0$ (2) $x + y + 4 = 0$
(3) $x - 2y + 16 = 0$ (4) $2x - y + 2 = 0$

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Q30

The tangents to the curve $y = (x - 2)^2 - 1$ at its points of intersection with the line $x - y = 3$, intersect at the point :

- (1) $\left(\frac{5}{2}, 1\right)$ (2) $\left(-\frac{5}{2}, -1\right)$
(3) $\left(\frac{5}{2}, -1\right)$ (4) $\left(-\frac{5}{2}, 1\right)$

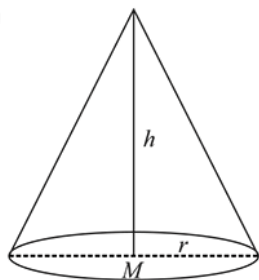
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Application of Derivatives - Solutions

Q1

(4)



$$h^2 + r^2 = l^2 = 9 \quad \dots(1)$$

Volume of cone

$$V = \frac{1}{3} \pi r^2 h \quad \dots(2)$$

From (1) and (2),

$$\Rightarrow V = \frac{1}{3} \pi (9 - h^2) h$$

$$\Rightarrow V = \frac{1}{3} \pi (9h - h^3) \Rightarrow \frac{dv}{dh} = \frac{1}{3} \pi (9 - 3h^2)$$

For maxima/minima,

$$\frac{dV}{dh} = 0 \Rightarrow \frac{1}{3} \pi (9 - 3h^2) = 0$$

$$\Rightarrow h = \pm \sqrt{3} \Rightarrow h = \sqrt{3} \quad (\because h > 0)$$

$$\text{Now, } \frac{d^2V}{dh^2} = \frac{1}{3} \pi (-6h)$$

$$\text{Here, } \left(\frac{d^2V}{dh^2} \right)_{\text{at } h=\sqrt{3}} < 0$$

Then, $h = \sqrt{3}$ is point of maxima

Hence, the required maximum volume is,

$$V = \frac{1}{3} \pi (9 - 3) \sqrt{3} = 2\sqrt{3} \pi$$

$$(1) \det(A) = \begin{vmatrix} -2 & 4+d & \sin \theta - 2 \\ 1 & \sin \theta + 2 & d \\ 5 & 2 \sin \theta - d & -\sin \theta + 2 + 2d \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - 2R_2 + R_1$ we get

$$\det(A) = \begin{vmatrix} -2 & 4+d & \sin \theta - 2 \\ 1 & \sin \theta + 2 & d \\ 1 & 0 & 0 \end{vmatrix}$$

$$= d(4+d) - (\sin^2 \theta - 4)$$

$$\Rightarrow \det(A) = d^2 + 4d + 4 - \sin^2 \theta = (d+2)^2 - \sin^2 \theta$$

Minimum value of $\det(A)$ is attained when $\sin^2 \theta = 1$

$$\therefore (d+2)^2 - 1 = 8 \Rightarrow (d+2)^2 = 9 \Rightarrow d+2$$

$$= \pm 3$$

$$\Rightarrow d = -5 \text{ or } 1$$

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Q3

Q2

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Application of Derivatives

$$(4) I = \int_a^b (x^4 - 2x^2) dx$$

$$\Rightarrow \frac{dI}{dx} = x^4 - 2x^2 = 0 \text{ (for minimum)}$$

$$\Rightarrow x = 0, \pm\sqrt{2}$$

Also,

$$I = \left[\frac{x^5}{5} - \frac{2x^3}{3} \right]_a^b$$

For $a = 0, b = \sqrt{2}$

$$I = \frac{8\sqrt{2}}{15}$$

For $a = -\sqrt{2}, b = 0$

$$I = \frac{-8\sqrt{2}}{15}$$

For $a = -\sqrt{2}, b = \sqrt{2}$

$$I = \frac{16\sqrt{2}}{15}$$

For $a = -\sqrt{2}, b = \sqrt{2}$

$$I = \frac{-16\sqrt{2}}{15}$$

$\therefore I$ is minimum when $(a, b) = (-\sqrt{2}, \sqrt{2})$

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Q4

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(2) The equation of curve $y = xe^{x^2}$

$$\Rightarrow \frac{dy}{dx} = e^{x^2} \cdot 1 + x \cdot e^{x^2} \cdot 2x$$

Since $(1, e)$ lies on the curve $y = xe^{x^2}$, then equation of tangent at $(1, e)$ is

$$y - e = \left(e^{x^2} (1 + 2x^2) \right)_{x=1} (x - 1)$$

$$y - e = 3e(x - 1)$$

$$3ex - y = 2e$$

So, equation of tangent to the curve passes through the point $\left(\frac{4}{3}, 2e \right)$

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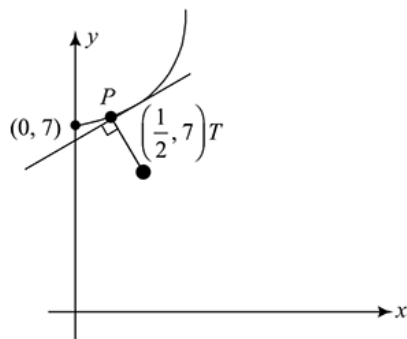
Q5

Application of Derivatives

(3) $f(x) = y = x^{3/2} + 7$

$$\Rightarrow \frac{dy}{dx} \Rightarrow \frac{3}{2}\sqrt{x} > 0$$

$$\Rightarrow f(x) \text{ is increasing function } \forall x > 0$$



Let $P(x_1, x_1^{3/2} + 7)$
 $m_{TP} = m_{at P} = -1$

$$\Rightarrow \left(\frac{x_1^{3/2}}{x_1 - \frac{1}{2}} \right) \times \frac{3}{2} x_1^{\frac{1}{2}} = -1$$

$$\Rightarrow -\frac{2}{3} = \frac{x_1^2}{x_1 - \frac{1}{2}}$$

$$\Rightarrow -3x_1^2 = 2x_1 - 1 \Rightarrow 3x_1^2 + 2x_1 - 1 = 0$$

$$\Rightarrow 3x_1^2 + 3x_1 - x_1 - 1 = 0$$

$$\Rightarrow 3x_1(x_1 + 1) - 1(x_1 + 1) = 0$$

$$\Rightarrow x_1 = \frac{1}{3} \quad (\because x_1 > 0) \Rightarrow P\left(\frac{1}{3}, 7 + \frac{1}{3\sqrt{3}}\right)$$

$$TP = \sqrt{\frac{1}{27} + \frac{1}{36}} = \frac{1}{6}\sqrt{\frac{7}{3}}$$

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Q6

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(3) Consider the function,

$$f(x) = 3x(x-3)^2 - 40$$

$$\text{Now } S = \{x \in R : x^2 + 30 \leq 11x\}$$

$$\text{So } x^2 - 11x + 30 \leq 0$$

$$\Rightarrow x \in [5, 6]$$

$\therefore f(x)$ will have maximum value for $x = 6$

The maximum value of function is,

$$f(6) = 3 \times 6 \times 3 \times 3 - 40 = 122.$$

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Q9

$$(3) A = \frac{x^m y^n}{(1+x^{2m})(1+y^{2n})} = \frac{1}{(x^{-m} + x^m)(y^{-n} + y^n)}$$

$$\frac{x^m + y^{-m}}{2} \geq (x^m \cdot x^{-m})^{\frac{1}{2}} \Rightarrow x^m + x^{-m} \geq 2$$

In the same way, $y^{-n} + y^n \geq 2$

Then, $(x^m + x^{-m})(y^{-n} + y^n) \geq 4$

$$\Rightarrow \frac{1}{(x^m + x^{-m})(y^{-n} + y^n)} \leq \frac{1}{4}$$

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Q10

Application of Derivatives

$$\begin{aligned}
 (1) f(x) &= \frac{x}{\sqrt{a^2+x^2}} - \frac{(d-x)}{\sqrt{b^2+(d-x)^2}} \\
 &= \frac{x}{\sqrt{a^2+x^2}} + \frac{(x-d)}{\sqrt{b^2+(x-d)^2}} \\
 f'(x) &= \frac{\sqrt{a^2+x^2} - \frac{x(2x)}{2\sqrt{a^2+x^2}}}{(a^2+x^2)} \\
 &\quad + \frac{\sqrt{b^2+(x-d)^2} - \frac{(x-d)2(x-d)}{2\sqrt{b^2+(x-d)^2}}}{(b^2+(x-d)^2)} \\
 &= \frac{a^2+x^2-x^2}{(a^2+x^2)^{3/2}} + \frac{b^2+(x-d)^2-(x-d)^2}{(b^2+(x-d)^2)^{3/2}} \\
 &= \frac{a^2}{(a^2+x^2)^{3/2}} + \frac{b^2}{(b^2+(x-d)^2)^{3/2}} > 0 \\
 \Rightarrow f'(x) &> 0, \forall x \in R \\
 \Rightarrow f(x) &\text{ is increasing function.} \\
 \text{Hence, } f(x) &\text{ is increasing function.}
 \end{aligned}$$

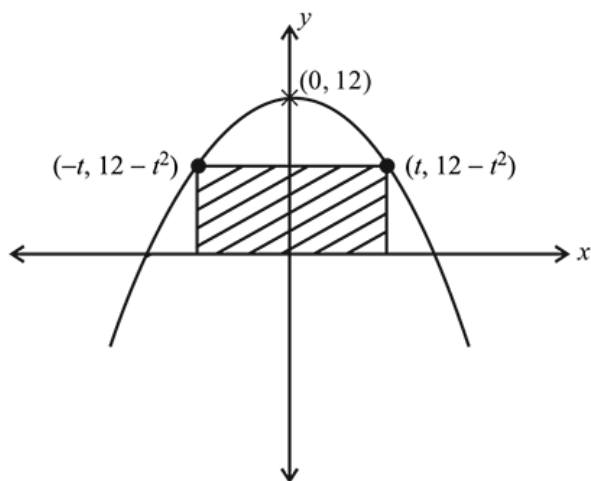
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Q11

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(3) Given, the equation of parabola is,

$$x^2 = 12 - y$$



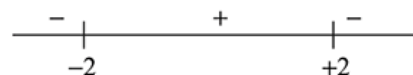
Area of the rectangle = $(2t)(12 - t^2)$

$$A = 24t - 2t^3$$

$$\frac{dA}{dt} = 24 - 6t^2$$

$$\text{Put } \frac{dA}{dt} = 0 \Rightarrow 24 - 6t^2 = 0$$

$$\Rightarrow t = \pm 2$$



At $t = 2$, area is maximum = $24(2) - 2(2)^3$

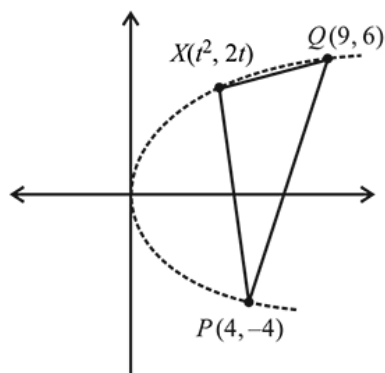
$$= 48 - 16 = 32 \text{ sq. units}$$

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Q12

Application of Derivatives

(2)



Parametric equations of the parabola $y^2 = 4x$ are,

$$x = t^2 \text{ and } y = 2t.$$

$$\text{Area } \Delta PXQ = \frac{1}{2} \begin{vmatrix} t^2 & 2t & 1 \\ 4 & -4 & 1 \\ 9 & 6 & 1 \end{vmatrix}$$

$$= -5t^2 + 5t + 30$$

$$= -5(t^2 - t - 6)$$

$$= -5 \left[\left(t - \frac{1}{2} \right)^2 - \frac{25}{4} \right]$$

For maximum area $t = \frac{1}{2}$

$$\therefore \text{maximum area} = 5 \left(\frac{25}{4} \right) = \frac{125}{4}$$

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Q13

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(1) Let, the functions is,

$$f(\theta) = 3 \cos \theta + 5 \sin \theta \cdot \cos \frac{\pi}{6} - 5 \sin \frac{\pi}{6} \cos \theta$$

$$= 3 \cos \theta + 5 \times \frac{\sqrt{3}}{2} \sin \theta - 5 \times \frac{1}{2} \cos \theta$$

$$= \left(3 - \frac{5}{2} \right) \cos \theta + 5 \times \frac{\sqrt{3}}{2} \sin \theta$$

$$= \frac{1}{2} \cos \theta + \frac{5\sqrt{3}}{2} \sin \theta$$

$$\max f(\theta) = \sqrt{\frac{1}{4} + \frac{25}{4}} \times 3 = \sqrt{\frac{76}{4}} = \sqrt{19}$$

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Q14

Application of Derivatives

$$(2) \because \text{Slope of the tangent} = \frac{x^2 - 2y}{x}$$

$$\therefore \frac{dy}{dx} = \frac{x^2 - 2y}{x}$$

$$\frac{dy}{dx} + \frac{2}{x}y = x$$

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

Solution of equation

$$y \cdot x^2 = \int x \cdot x^2 dx$$

$$x^2 y = \frac{x^4}{4} + C$$

$\therefore$ curve passes through point $(1, -2)$

$$(1)^2(-2) = \frac{1^4}{4} + C$$

$$\Rightarrow C = \frac{-9}{4}$$

Then, equation of curve

$$y = \frac{x^2}{4} - \frac{9}{4x^2}$$

Since, above curve satisfies the point.

Hence, the curve passes through $(\sqrt{3}, 0)$.

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Q15

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(2) $\because$ tangent to the given curve is parallel to line $2y = 4x + 1$

$\therefore$ Slope of tangent (m) = 2

Then, the equation of tangent will be of the form

$$y = 2x + c \quad \dots(1)$$

$\because$ Line (1) and curve $y = x^2 - 5x + 5$ has only one point of intersection.

$$\therefore 2x + c = x^2 - 5x + 5$$

$$x^2 - 7x + (5 - c) = 0$$

$$\therefore D = 49 - 4(5 - c) = 0$$

$$\Rightarrow c = -\frac{29}{4}$$

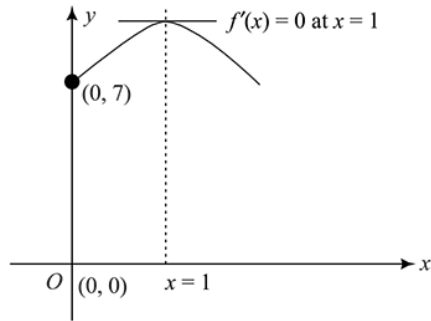
Hence, the equation of tangent: $y = 2x - \frac{29}{4}$

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Q16

Application of Derivatives

(3) $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7, f(0) = 7$



$$\Rightarrow f'(x) = 3x^2 - 6(a-2)x + 3a$$

$$f'(1) = 0$$

$$\Rightarrow 1 - 2a + 4 + a = 0$$

$$\Rightarrow a = 5$$

Then, $f(x) = x^3 - 9x^2 + 15x + 7$

Now,

$$\frac{f(x) - 14}{(x-1)^2} = 0$$

$$\Rightarrow \frac{x^3 - 9x^2 + 15x + 7 - 14}{(x-1)^2} = 0$$

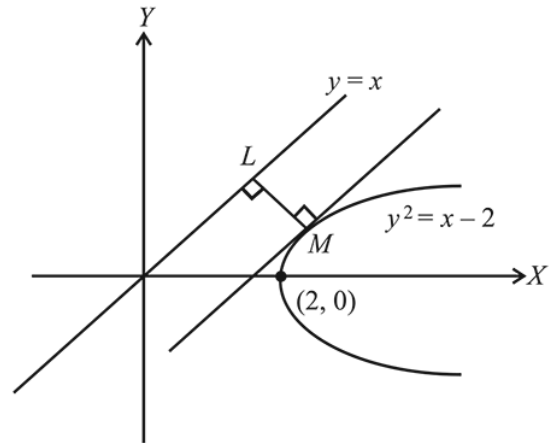
$$\Rightarrow \frac{(x-1)^2(x-7)}{(x-1)^2} = 0 \Rightarrow x = 7$$

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Q17

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- (3) The shortest distance between line $y = x$ and parabola $y^2 = x - 2$ is the distance LM between line $y = x$ and tangent of parabola having slope 1.



Let equation of tangent of parabola having slope 1 is,

$$y = m(x - 2) + \frac{a}{m}$$

Here $m = 1$ and $a = \frac{1}{4}$

$\therefore$ equation of tangent is: $y = x - \frac{7}{4}$

Distance between the line $y - x = 0$ and $y - x + \frac{7}{4} = 0$

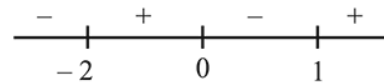
$$= \left| \frac{\frac{7}{4} - 0}{\sqrt{1^2 + 1^2}} \right| = \frac{7}{4\sqrt{2}}$$

8 April Morning

Q18

(3) $f(x) = 9x^4 + 12x^3 - 36x^2 + 25$

$$f'(x) = 36[x^3 + x^2 - 2x] = 36x(x-1)(x+2)$$



Here at -2 & 1 , $f'(x)$ changes from negative value to positive value.

$\Rightarrow -2$ & 1 are local minimum points. At 0 , $f'(x)$ changes from positive value to negative value.

$\Rightarrow 0$ is the local maximum point.

Hence, $S_1 = \{-2, 1\}$ and $S_2 = \{0\}$

8 April Morning

Q19

Application of Derivatives

(3) $\phi(x) = f(x) + f(2-x)$

Now, differentiate w.r.t. x ,

$$\phi'(x) = f'(x) - f'(2-x)$$

For $\phi(x)$ to be increasing $\phi'(x) > 0$

$$\Rightarrow f'(x) - f'(2-x) > 0$$

$$\Rightarrow f'(x) > f'(2-x)$$

But $f''(x) > 0 \Rightarrow f'(x)$ is an increasing function

$$\text{Then, } f'(x) > f'(2-x) > 0$$

$$\Rightarrow x > 2-x$$

$$\Rightarrow x > 1$$

Hence, $\phi(x)$ is increasing on $(1, 2)$ and decreasing on $(0, 1)$.

8 April Morning

Q20

(1) Given $\frac{dy}{dx} = \frac{2y}{x^2}$

Integrating both sides, $\int \frac{dy}{y} = 2 \int \frac{dx}{x^2}$

$$\Rightarrow \ln|y| = -\frac{2}{x} + C \quad \dots(i)$$

Equation (i) passes through the centre of the circle

$$x^2 + y^2 - 2x - 2y = 0, \text{ i.e., } (1, 1)$$

$$\therefore C = 2$$

$$\text{Now, } \ln|y| = -\frac{2}{x} + 2$$

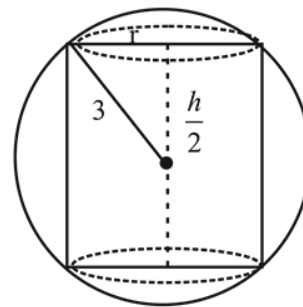
$$x \ln|y| = -2(1-x) \Rightarrow x \ln|y| = 2(x-1)$$

8 April Evening

Q21

JEE Mains 2019 Chapter wise Question Bank

- (3) Let radius of base and height of cylinder be r and h respectively.



$$\therefore r^2 + \frac{h^2}{4} = 9 \quad \dots(i)$$

Now, volume of cylinder, $V = \pi r^2 h$

Substitute the value of r^2 from equation (i),

$$V = \pi h \left(9 - \frac{h^2}{4} \right) \Rightarrow V = 9\pi h - \frac{\pi}{4} h^3$$

Differentiating w.r.t. h ,

$$\frac{dV}{dh} = 9\pi - \frac{3}{4}\pi h^2$$

For maxima/minima,

$$\frac{dV}{dh} = 0 \Rightarrow h = \sqrt{12}$$

$$\text{and } \frac{d^2V}{dh^2} = -\frac{3}{2}\pi h$$

$$\therefore \left(\frac{d^2V}{dh^2} \right)_{h=\sqrt{12}} < 0$$

$\Rightarrow$ Volume is maximum when $h = 2\sqrt{3}$

8 April Evening

Q22

Application of Derivatives

- (4) Since, function $f(x)$ have local extreem points at $x = -1, 0, 1$. Then

$$f(x) = K(x+1)x(x-1)$$

$$= K(x^3 - x)$$

$$\Rightarrow f(x) = K\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + C \quad (\text{using integration})$$

$$\Rightarrow f(0) = C$$

$$\because f(x) = f(0) \Rightarrow K\left(\frac{x^4}{4} - \frac{x^2}{2}\right) = 0$$

$$\Rightarrow \frac{x^2}{2}\left(\frac{x^2}{2} - 1\right) = 0$$

$$\Rightarrow x = 0, 0, \sqrt{2}, -\sqrt{2}$$

$$\therefore S = \{0, -\sqrt{2}, \sqrt{2}\}$$

9 April Morning

Q23

(4) $y = x^3 + ax - b$

Since, the point $(1, -5)$ lies on the curve.

$$\Rightarrow 1 + a - b = -5$$

$$\Rightarrow a - b = -6 \quad \dots(1)$$

Now, $\frac{dy}{dx} = 3x^2 + a$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{\text{at } x=1} = 3 + a$$

Since, required line is perpendicular to $y = x - 4$, then slope of tangent at the point $P(1, -5) = -1$

$$\therefore 3 + a = -1$$

$$\Rightarrow a = -4$$

$$\Rightarrow b = 2$$

$$\therefore \text{the equation of the curve is } y = x^3 - 4x - 2$$

$$\Rightarrow (2, -2) \text{ lies on the curve}$$

9 April Morning

Q24

JEE Mains 2019 Chapter wise Question Bank

(4) $y = f(x) = x^3 - x^2 - 2x$

$$\Rightarrow \frac{dy}{dx} = 3x^2 - 2x - 2$$

$$f(1) = 1 - 1 - 2 = -2, \quad f(-1) = -1 - 1 + 2 = 0$$

Since the tangent to the curve is parallel to the line segment joining the points $(1, -2)$ and $(-1, 0)$

And their slopes are equal.

$$\Rightarrow 3x^2 - 2x - 2 = \frac{-2 - 0}{2}$$

$$\Rightarrow x = 1, \frac{-1}{3}$$

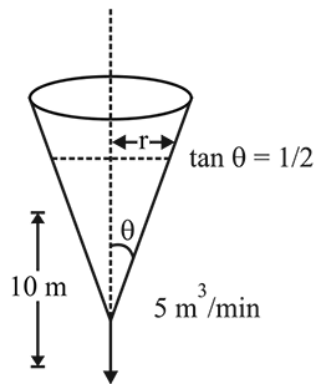
$$\text{Hence, the required set } S = \left\{\frac{-1}{3}, 1\right\}$$

9 April Morning

Q25

Application of Derivatives

(4)



Given that water is poured into the tank at a constant rate of $5 \text{ m}^3/\text{min}$.

$$\therefore \frac{dV}{dt} = 5 \text{ m}^3/\text{min}$$

Volume of the tank is,

$$V = \frac{1}{3} \pi r^2 h \quad \dots(i)$$

where r is radius and h is height at any time.

By the diagram,

$$\tan \theta = \frac{r}{h} = \frac{1}{2}$$

$$\Rightarrow h = 2r \Rightarrow \frac{dh}{dt} = \frac{2dr}{dt} \quad \dots(ii)$$

Differentiate eq. (i) w.r.t. 't', we get

$$\frac{dV}{dt} = \frac{1}{3} \left(\pi 2r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt} \right)$$

Putting $h = 10$, $r = 5$ and $\frac{dV}{dt} = 5$ in the above equation.

$$5 = \frac{75\pi}{3} \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{5\pi} \text{ m/min.}$$

9 April Evening

Q26

JEE Mains 2019 Chapter wise Question Bank

(2) Given functions are, $f(x) = e^x - x$ and $g(x) = x^2 - x$

$$f(g(x)) = e^{(x^2-x)} - (x^2-x)$$

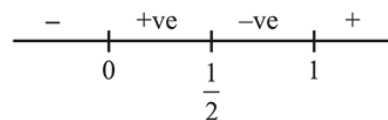
Given $f(g(x))$ is increasing function.

$$\therefore (f(g(x)))' = e^{(x^2-x)} \times (2x-1) - 2x+1$$

$$= (2x-1)e^{(x^2-x)} + 1 - 2x = (2x-1)[e^{(x^2-x)} - 1] \geq 0$$

For $(f(g(x)))' \geq 0$,

$(2x-1)$ & $[e^{(x^2-x)} - 1]$ are either both positive or negative



$$x \in \left[0, \frac{1}{2}\right] \cup [1, \infty)$$

10 April Morning

Q27

(1) Given curve is, $y = \frac{x}{x^2-3}$

$$\Rightarrow \frac{dy}{dx} = \frac{(x^2-3) - x(2x)}{(x^2-3)^2} = \frac{-x^2-3}{(x^2-3)^2}$$

$$\left. \frac{dy}{dx} \right|_{(\alpha, \beta)} = \frac{\alpha^2-3}{(\alpha^2-3)^2} = -\frac{2}{6} = -\frac{1}{3}$$

$$3(\alpha^2+3) = (\alpha^2-3)^2 \Rightarrow \alpha^2 = 9$$

$$\text{And, } \beta = \frac{x}{\alpha^2-3} \Rightarrow \alpha^2-3 = \frac{\alpha}{\beta} \Rightarrow \frac{\alpha}{\beta} = 6$$

$$\Rightarrow \alpha = \pm 3, \beta = \pm \frac{1}{2}$$

These values of α and β satisfies $|6\alpha + 2\beta| = 19$

10 April Evening

Q28

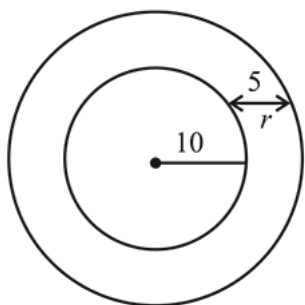
Application of Derivatives

- (1) Given that ice melts at a rate of $50 \text{ cm}^3/\text{min}$.

$$\therefore \frac{dV_{\text{ice}}}{dt} = 50$$

$$V_{\text{ice}} = \frac{4}{3}\pi(10+r)^3 - \frac{4}{3}\pi(10)^3$$

$$\Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi 3(10+r)^2 \frac{dr}{dt} = 4\pi(10+r)^2 \frac{dr}{dt}$$



Substitute $r = 5$,

$$50 = 4\pi(225) \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{50}{4\pi(225)} = \frac{1}{18\pi} \text{ cm/min}$$

10 April Evening

Q29

- (2) $a_6 = a + 5d = 2$

Here, a is first term of A.P and d is common difference

$$\text{Let } A = a_1 a_4 a_5 = a(a + 3d)(a + 4d)$$

$$= a(2 - 2d)(2 - d)$$

$$A = (2 - 5d)(4 - 6d + 2d^2)$$

$$\text{By } \frac{dA}{dd} = 0$$

$$\Rightarrow (2 - 5d)(-6 + 4d) + (4 - 6d + 2d^2)(-5) = 0$$

$$\Rightarrow -15d^2 + 34d - 16 = 0 \Rightarrow d = \frac{8}{5}, \frac{2}{3}$$

$$\text{For } d = \frac{8}{5}, \frac{d^2 A}{dd^2} < 0. \text{ Hence } d = \frac{8}{5}$$

10 April Evening

Q30

JEE Mains 2019 Chapter wise Question Bank

- (2) Given function $f(x) = x\sqrt{kx - x^2} = \sqrt{kx^3 - x^4}$

Differentiating w. r. t. x ,

$$f'(x) = \frac{(3kx^2 - 4x^3)}{2\sqrt{kx^3 - x^4}} \geq 0 \text{ for } x \in [0, 3]$$

$[\because f(x)$ is increasing in $[0, 3]]$

$$\Rightarrow 3k - 4x \geq 0 \Rightarrow 3k \geq 4x$$

$$\text{i.e., } 3k \geq 4x \text{ for } x \in [0, 3]$$

$$\therefore k \geq 4 \text{ i.e., } m = 4$$

$$\text{Putting } k = 4 \text{ in the function, } f(x) = x\sqrt{4x - x^2}$$

$$\text{For max. value, } f'(x) = 0$$

$$\text{i.e. } \frac{12x^2 - 4x^3}{2\sqrt{4x^3 - x^4}} = 0 \Rightarrow x = 3$$

$$\therefore y = 3\sqrt{3} \text{ i.e., } M = 3\sqrt{3}$$

12 April Morning

Q31

- (1) Given curves, $y^2 = 16x$ and $xy = -4$

Equation of tangent to the given parabola;

$$y = mx + \frac{4}{m}$$

$\therefore$ This is common tangent.

$$\text{So, put } y = mx + \frac{4}{m} \text{ in } xy = -4.$$

$$x\left(mx + \frac{4}{m}\right) + 4 = 0 \Rightarrow mx^2 + \frac{4}{m}x + 4 = 0$$

$$D = 0 \Rightarrow \frac{16}{m^2} = 16m \Rightarrow m^3 = 1 \Rightarrow m = 1$$

$\therefore$ equation of common tangent is $y = x + 4$

12 April Evening

Q32

- (3) Tangent to the curve $y = (x - 2)^2 - 1$ at any point (h, k) is,

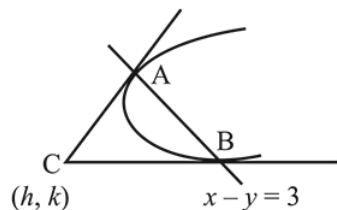
$$\Rightarrow \frac{1}{2}(y + k) = (x - 2)(h - 2) - 1$$

$$\Rightarrow \frac{y + k}{2} = xh - 2x - 2h + 3$$

$$\Rightarrow (2h - 4)x - y - 4h + 6 - k = 0$$

Given line, $x - y - 3 = 0$

$$\Rightarrow \frac{2h - 4}{1} = \frac{4h - 6 + k}{3} = 1$$



$$\Rightarrow h = \frac{5}{2}, k = -1$$

12 April Evening



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